Do Now · retrieval

Trees and binary search trees: implementation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does a null Next mean in a singly linked list?

Q2

In a BST, where does a smaller key go?

Q3

What is the base case of a recursive insertion reaching an empty child?

Q4

Insert 10,20,30 in that order into an empty unbalanced BST. What shape results?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find the starting node — The root is 40, so compare 25 with 40 and move left.

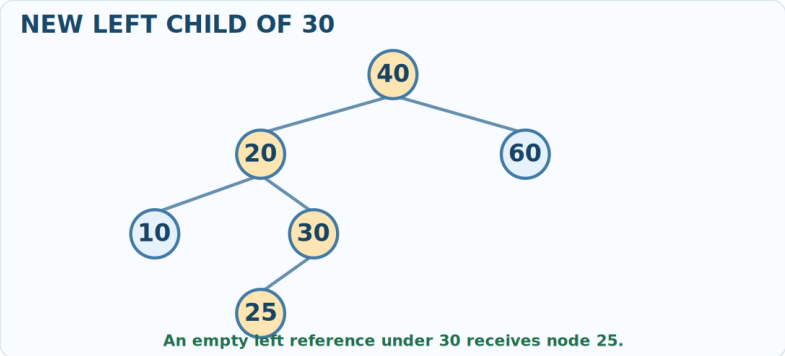
Core idea 2

Repeat comparisons — 25>20 goes right to 30; 25<30 goes left.

Core idea 3

Insert at a null child — Create node 25 at the empty left-child position of 30. This implementation ignores duplicate keys: when key equals node.Key, neither subtree changes.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

binary search tree — Binary tree in which lower keys occur to the left and higher to the right.

root — Topmost tree node.

leaf — A node with no children.

in-order traversal — Visit left subtree, node, then right subtree.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

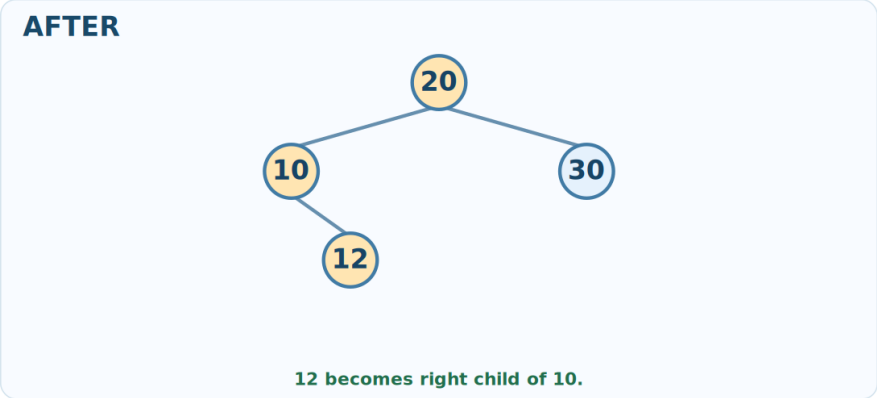
Problem

Insert 12 into the BST 20 (root), 10 (left), 30 (right).

Follow the worked method

1. Predict first — Compare 12 with 20, then 10.
2. Model the reasoning — Insert 12 as the right child of 10, preserving the BST property.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
35>25 leads right from the root.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Insert 12 into BST root 15, left 9 and right 20. Identify its parent and position.

Question 2

Insert 50,30,70,20,40,35 into an empty BST. State parent and side of 35, and the in-order visit sequence.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

In BST 40 with children 20 and 60, and 20 with children 10 and 30, delete leaf 10. Explain a second case in which deletion is more involved.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

In-order traversal of a BST with distinct keys visits values in ascending order.

Use key vocabulary from memory

1. Explain binary search tree in your own words.

2. Explain root in your own words.

3. Explain leaf in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.